



Summary

- ▶ We have seen three possible meanings for rotation matrices

Geometric Meanings of a Rotation Matrix

1. It describes the mutual orientation between two coordinated frames. Its column vectors are the direction cosines of the axes of the rotated frame with respect to the original one.
2. It represents the coordinate transformation between two rotated frames
3. It is the operator that enables rotations within the same coordinate frame



Composition of Rotation Matrices

Let us start with an example

- ▶ Consider three different frames sharing the same origin:
 $O - x_0y_0z_0$, $O - x_1y_1z_1$, $O - x_2y_2z_2$
- ▶ Consider a point P with coordinates p^0 , p^1 and p^2 in the three frames
- ▶ Let R_i^j denote the rotation matrix of Frame i with respect to Frame j .
- ▶ We have:

$$p^1 = R_2^1 p^2$$

$$p^0 = R_1^0 p^1$$

$$p^0 = R_2^0 p^2$$

- ▶ By simple substitution and comparison

$$p^0 = R_1^0 R_2^1 p^2$$

$$R_2^0 = R_1^0 R_2^1$$

$$p^1 = R_2^1 p^2 \rightarrow$$





Composition of Rotation Matrices

We can interpret the relation $R_2^0 = R_1^0 R_2^1$ as obtained in the two following steps

1. First rotate $O - x_0y_0z_0$ so as to align it with frame $O - x_1y_1z_1$ (matrix R_1^0)
2. Then rotate again so as to align frame $O - x_1y_1z_1$ with $O - x_2y_2z_2$ (matrix R_2^1)

Generally speaking:

Composition of Rotations

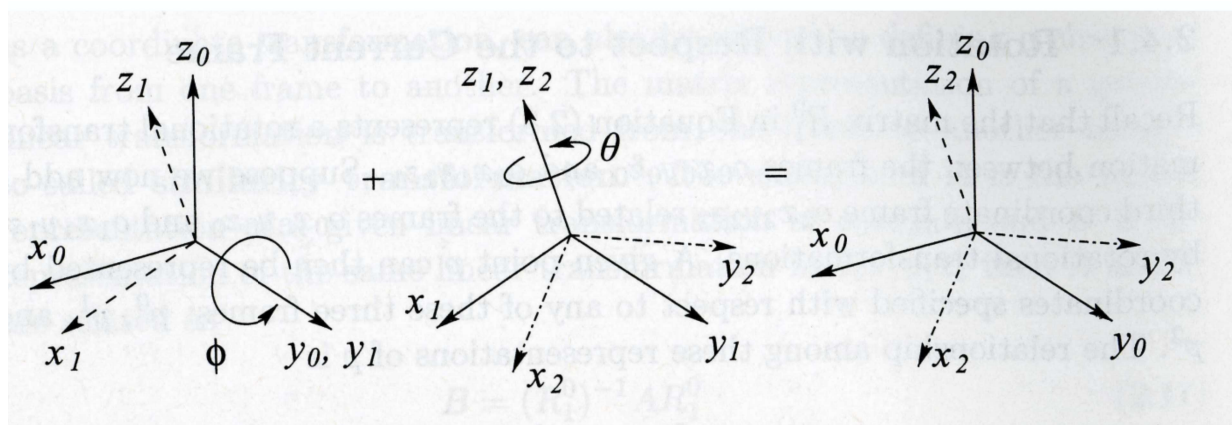
A rotation can be expressed as a sequence of partial rotations, each one defined w.r.t. the previous one. The frame from which the rotation occurs is termed *current frame*. The sequence is obtained by post-multiplication of the matrices associated with each rotation.

In view of the properties of rotation matrices, we have:

$$R_i^j = (R_j^i)^{-1} = (R_j^i)^T$$

Composition of Rotation Matrices - Example

An example of a sequence of rotations about axes in *the current frame* is below



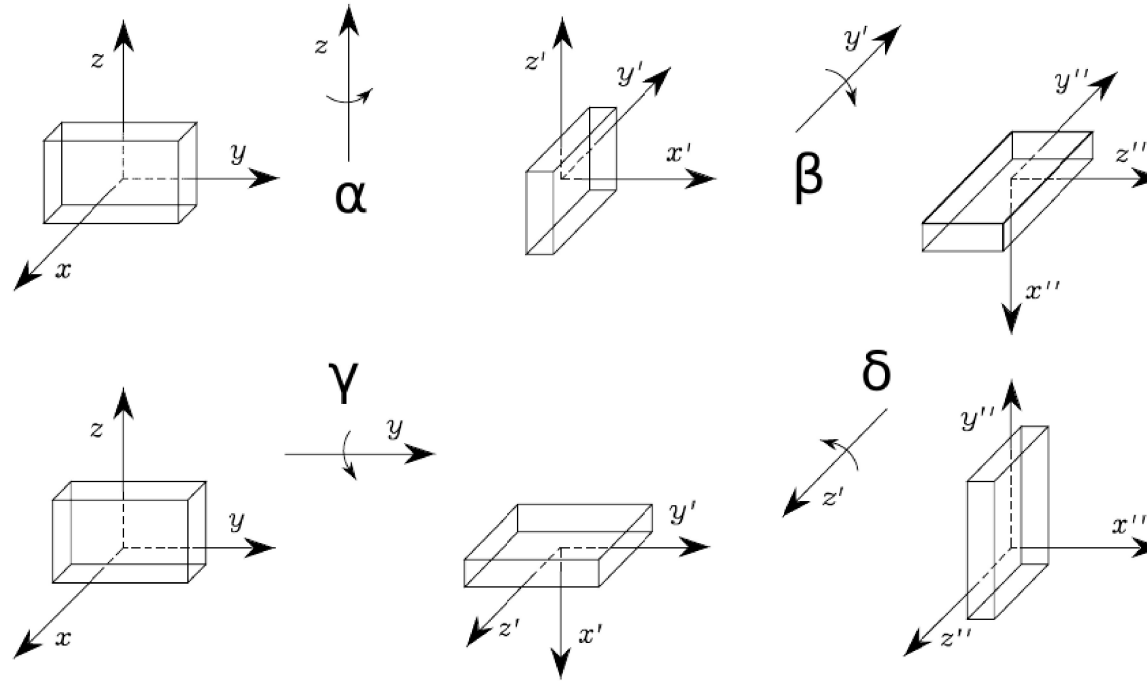
- ▶ We first rotate by ϕ about the current y -axis
- ▶ Then we rotate by θ about the current z -axis

Result:

$$R_2^0 = R_y(\phi)R_z(\theta).$$

Composition of Rotation Matrices - Another example

An example of a sequence of rotations about axes in *the current frame* is below



The resulting rotation matrix is:

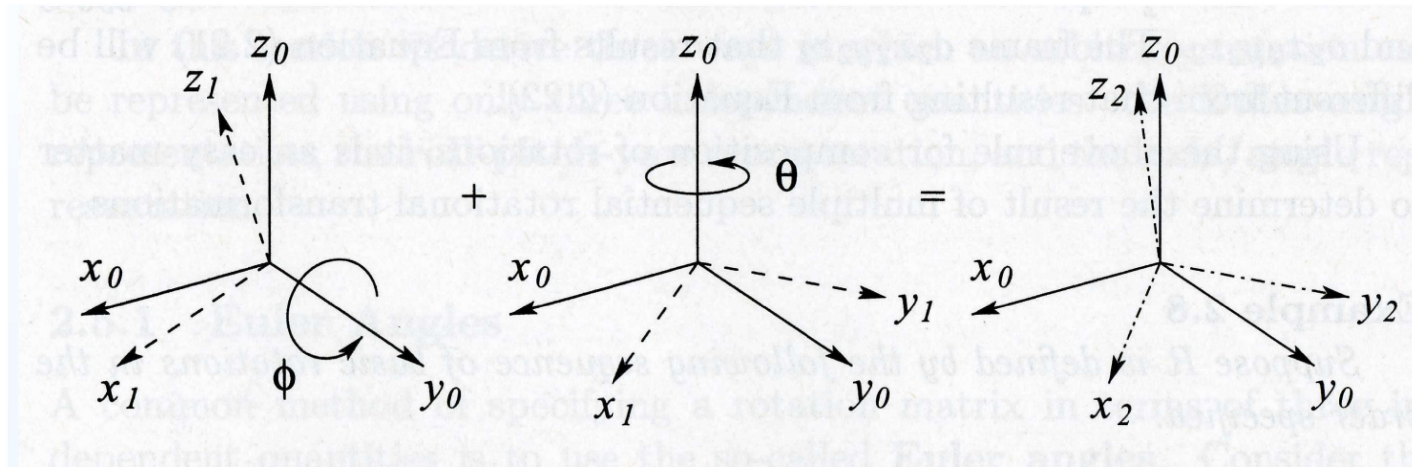
$$R_4^0 = R_z(\alpha)R_y(\beta)R_y(\gamma)R_z(\delta).$$

The order is absolutely important!

$$R_z(\alpha)R_y(\beta)R_y(\gamma)R_z(\delta) \neq R_y(\gamma)R_z(\delta)R_z(\alpha)R_y(\beta).$$

Rotation about fixed axes

- What if we want to specify the sequence of rotations about fixed axes
- An example of a sequence of rotations *in a fixed frame* is below



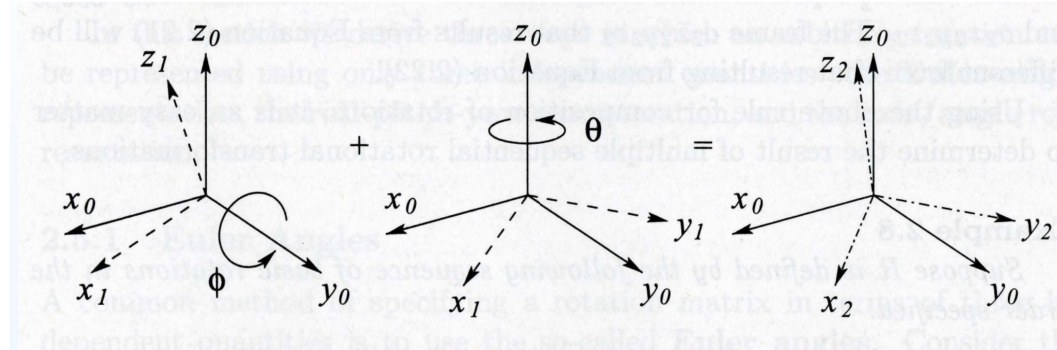


Rotation about fixed axes

- ▶ In order to see how rotations can be composed in a fixed axis, let us use a simple fact
- ▶ Consider two rotated frame $O - x_0y_0z_0$ and $O - x_1y_1z_1$.
- ▶ For a generic point P we have: $p^0 = R_1^0 p^1$
- ▶ Consider a generic linear application A defined in the $O - x_0y_0z_0$ frame.
- ▶ For a generic point P its transformed point we have:

$$\begin{aligned} q^0 &= Ap^0 \\ q^1 &= R_0^1 Ap^0 = R_0^1 AR_1^0 p^1 = (R_1^0)^{-1} AR_1^0 p^1 \end{aligned}$$

Rotation about fixed axes - back to the example



We have two rotations

- ▶ The first rotation $R_1^0 = R_y(\phi)$ around y_0 -axis
- ▶ The rotation $\bar{R}_2^1$, which happens around the z_0 -axis (not z_1 as in the case of the current frame)

$$R_2^0 = R_1^0 \bar{R}_2^1.$$

We know how the latter is done in the current axes. So we can use the result above

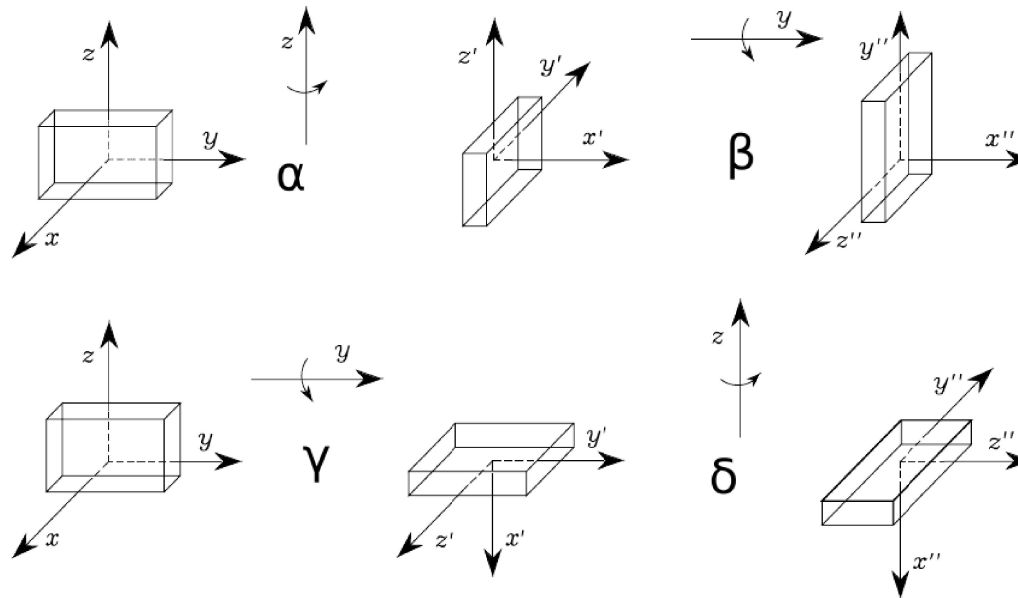
$$\bar{R}_2^1 = (R_1^0)^{-1} R_z(\theta) R_1^0,$$

which leads to

$$R_2^0 = R_1^0 \bar{R}_2^1 = R_1^0 (R_1^0)^{-1} R_z(\theta) R_1^0 = R_z(\theta) R_y(\phi)$$

Rotation about fixed axes - back to the example

We can conclude that when the composition of rotation happens with fixed axes we have to premultiply instead of postmultiplying (as is the case for rotations in the current frame). Consider the example:



We have

$$R_4^0 = R_z(\delta)R_y(\gamma)R_y(\beta)R_z(\alpha)$$



Rotation about current axes - Example

Consider the rotation composed as follows:

1. A rotation of θ about the current axis x
2. A rotation of ϕ around the current axis z

The result is:

$$R_2^0 = R_x(\theta)R_z(\phi).$$

A point expressed p^2 in the frame $0 - x_2y_2z_2$ corresponds to $p^0 = R_2^0 p^2$ in the initial frame.

A more articulated Example

Consider the rotation composed as follows:

1. A rotation of θ about the current axis x
2. A rotation of ϕ around the current axis z
3. A rotation of α about the *fixed* axis z_0

Following the same procedure as above, we have:

The result is:

$$R_3^0 = R_1^0 R_2^1 \bar{R}_3^2 = R_x(\theta) R_z(\phi) \bar{R}_3^2.$$

where $\bar{R}_3^2$ denotes the notation with respect to the initial fixed axes. The effect of the rotation about the fixed frame in the current frame

$O - x_2 y_2 z_2$ is given by

$$\bar{R}_3^2 = (R_1^0 R_2^1)^{-1} R_z(\alpha) R_1^0 R_2^1 = (R_x(\theta) R_z(\phi))^{-1} R_z(\alpha) R_x(\theta) R_z(\phi)$$

which leads us to

$$R_3^0 = R_z(\alpha) R_x(\theta) R_z(\phi).$$

The Ultimate Example

Consider the rotation composed as follows:

1. A rotation of θ about the current axis x
2. A rotation of ϕ around the current axis z
3. A rotation of α about the *fixed* axis z_0
4. A rotation of β around the current axis y
5. A rotation of δ about the fixed axis x_0

Following the same procedure that we have seen before, we come up with:

$$R_r^0 = R_x(\delta)R_z(\alpha)R_x(\theta)R_z(\phi)R_y(\beta).$$

Please verify yourself paper and pencil...



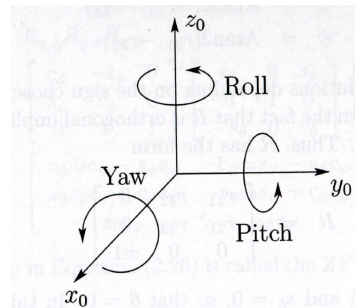
Parametrisation of Rotation Matrices

- ▶ A 3D rotation matrix has 9 elements, but in fact a rotation is completely specified by three different parameters.
- ▶ Generally speaking, a rotation is an element of $SO(m)$ that is completely specified by $m(m - 1)/2$ parameters
 - ▶ 1 parameters for $SO(2)$
 - ▶ 3 parameters for $SO(3)$
- ▶ A 3D rotation is completely specified by three different rotations about different axis such that two adjacent rotations are not about the same axis.
- ▶ As a result there are 12 parametrisations with three rotations:

3×	Any of the three axes is ok for the first rotation
2×	Rule out the axes used for the first rotation
2 =	Rule out the axes used for the second rotation
12	

Euler Angles

- ▶ Euler angles are the angles related to three different rotations about the current axis
- ▶ Very popular is the use of the XYZ rotation about fixed axis, which is known in the aeronautical world as roll-pitch-yaw because they denote the change of attitude of an aircraft with respect to three fixed axis.



- ▶ Since the rotation is about fixed axis we have to premultiply:

$$R = R_z(\phi)R_y(\theta)R_x(\psi)$$

- ▶ ϕ is the roll angle, θ is the pitch angle, ψ is the yaw angle.

RPY Angles

- ▶ Introduce the notation $c_\alpha = \cos(\alpha)$, $s_\alpha = \sin(\alpha)$.
- ▶ It is easy to see that

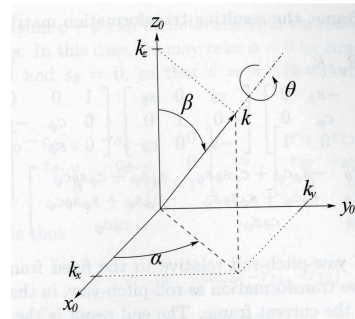
$$R = \begin{bmatrix} c_\phi c_\theta & c_\phi s_\theta s_\psi - s_\phi c_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ s_\phi c_\theta & s_\phi s_\theta s_\psi + c_\phi c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi \\ -s_\theta & c_\theta s_\psi & c_\theta c_\psi \end{bmatrix}$$

- ▶ given a Rotation matrix $R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$ we can solve the inverse problem (try it as an exercise) and find two solutions:

$$\begin{aligned} \theta \in (-\pi/2, \pi/2) &\rightarrow \begin{cases} \phi = \text{atan2}(r_{21}, r_{11}) \\ \theta = \text{atan2}\left(-r_{31}, \sqrt{r_{32}^2 + r_{33}^2}\right) \\ \psi = \text{atan2}(r_{32}, r_{33}) \end{cases} \\ \theta \in (\pi/2, 3\pi/2) &\rightarrow \begin{cases} \phi = \text{atan2}(-r_{21}, -r_{11}) \\ \theta = \text{atan2}\left(-r_{31}, -\sqrt{r_{32}^2 + r_{33}^2}\right) \\ \psi = \text{atan2}(-r_{32}, -r_{33}) \end{cases} \end{aligned}$$

Axis and rotation

- We can adopt a redundant parametrisation using an axis ($k = [k_x, k_y, k_z]$ with $|k| = 1$) and an angle (θ): four parameters overall.



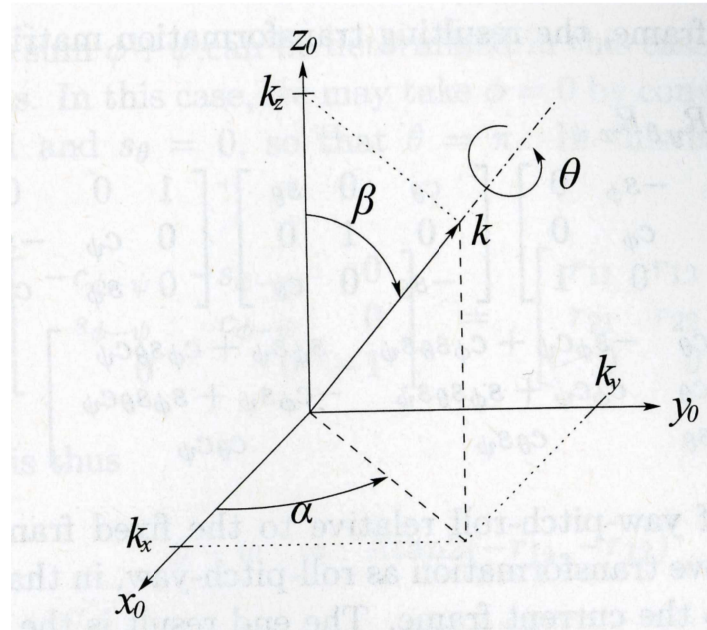
- The Rotation $R_k(\theta)$ can be found by transforming the coordinates
- We can find a new frame $O - x_1y_1z_1$, with z_1 coincident with k : we rotate $O - x_0y_0z_0$ by α about z and of β around y
- Clearly $R_1^0 = R_z(\alpha)R_y(\beta)$. If a transformation A (e.g., a rotation) is expressed in the frame $O - x_1y_1z_1$: $q^1 = Ap^1$, we know that

$$q^0 = (R_1^0)^{-1}AR_1^0p^0 = R_z(\alpha)R_y(\beta)A(R_z(\alpha)R_y(\beta))^{-1}$$

which leads to

$$R_k(\theta) = R_z(\alpha)R_y(\beta)R_z(\theta)R_y(-\beta)R_z(-\alpha)$$

Axis and rotation



With some trigonometry, we can find the α and β angles

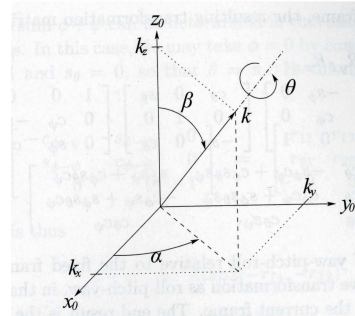
$$\sin(\alpha) = \frac{k_y}{\sqrt{k_x^2 + k_y^2}} \quad \cos(\alpha) = \frac{k_x}{\sqrt{k_x^2 + k_y^2}}$$

$$\sin(\beta) = \frac{\sqrt{k_x^2 + k_y^2}}{\sqrt{k_x^2 + k_y^2 + k_z^2}} \quad \cos(\beta) = \frac{k_z}{\sqrt{k_x^2 + k_y^2 + k_z^2}}$$

and

$$R_k(\theta) = \begin{bmatrix} k_x^2(1 - c_\theta) + c_\theta & k_x k_y(1 - c_\theta) - k_z s_\theta & k_x k_z(1 - c_\theta) + k_y s_\theta \\ k_x k_y(1 - c_\theta) + k_z s_\theta & k_y^2(1 - c_\theta) + c_\theta & k_y k_z(1 - c_\theta) - k_x s_\theta \\ k_x k_z(1 - c_\theta) - k_y s_\theta & k_y k_z(1 - c_\theta) + k_x s_\theta & k_z^2(1 - c_\theta) + c_\theta \end{bmatrix}$$

Axis and rotation



- ▶ We have specified a rotation matrix by four parameters (k and θ), but there is also a constraint ($k_x^2 + k_y^2 + k_z^2 = 1$). So the free parameters are actually three.
- ▶ From

$$R_k(\theta) = \begin{bmatrix} k_x^2(1 - c_\theta) + c_\theta & k_x k_y(1 - c_\theta) - k_z s_\theta & k_x k_z(1 - c_\theta) + k_y s_\theta \\ k_x k_y(1 - c_\theta) + k_z s_\theta & k_y^2(1 - c_\theta) + c_\theta & k_y k_z(1 - c_\theta) - k_x s_\theta \\ k_x k_z(1 - c_\theta) - k_y s_\theta & k_y k_z(1 - c_\theta) + k_x s_\theta & k_z^2(1 - c_\theta) + c_\theta \end{bmatrix}$$

We can see that $R_{-k}(-\theta) = R_k(\theta)$.

- ▶ If we invert axis and angle we get the same rotation

Sadly....

With axis and rotation, the Rotation Matrix is not unique and the representation is not invertible



Quaternions

- ▶ We can solve the invertibility and non-uniqueness problems of the axis and rotation approach by using quaternions
- ▶ A quaternion $Q = (\eta, \epsilon)$ is defined by four parameters a scalar η and a vector ϵ such that

$$\eta = \cos(\theta/2)$$

$$\epsilon = \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \end{bmatrix} = \sin(\theta/2) \mathbf{k}$$

where $\mathbf{k}$ is a unit vector.

- ▶ The four parameters are constrained

$$\eta^2 + \epsilon_x^2 + \epsilon_y^2 + \epsilon_z^2 = 1.$$

Quaternions

- ▶ Applying the same sequence of steps of axis and rotation, along with the trigonometric relations associated with the quaternion, which leads to

$$R(Q) = \begin{bmatrix} 2(\eta^2 + \epsilon_x^2) - 1 & 2(\epsilon_x \epsilon_y - \eta \epsilon_z) & 2(\epsilon_x \epsilon_z + \eta \epsilon_y) \\ 2(\epsilon_x \epsilon_y + \eta \epsilon_z) & 2(\eta^2 + \epsilon_y^2) - 1 & 2(\epsilon_y \epsilon_z - \eta \epsilon_x) \\ 2(\epsilon_x \epsilon_z - \eta \epsilon_y) & 2(\epsilon_y \epsilon_z + \eta \epsilon_x) & 2(\eta^2 + \epsilon_z^2) - 1 \end{bmatrix}$$

- ▶ Given a Rotation matrix $R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$ we find a *unique* solution for the inverse problem

$$\eta = \frac{1}{2} \sqrt{r_{11} + r_{22} + r_{33} + 1}$$

$$\epsilon = \frac{1}{2} \begin{bmatrix} \text{sgn}(r_{32} - r_{23}) \sqrt{r_{11} - r_{22} - r_{33} + 1} \\ \text{sgn}(r_{13} - r_{31}) \sqrt{r_{22} - r_{33} - r_{11} + 1} \\ \text{sgn}(r_{21} - r_{12}) \sqrt{r_{33} - r_{11} - r_{22} + 1} \end{bmatrix} \text{ where}$$

$$\text{sgn}(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases}$$

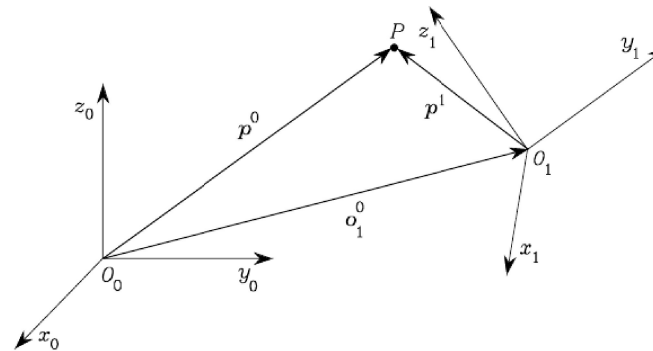
Quaternions

- The link between quaternions and rotation matrices is one to one. We can easily introduce operations on the quaternions that have the same results of the most used operations on rotation matrices.

Operation	Rotation matrices	Quaternions
Inversion	$R^{-1} = R^T$	$Q = (\eta, \epsilon) \rightarrow Q^{-1} \leftrightarrow (\eta, -\epsilon)$
Composition	$R_1, R_2 \rightarrow R = R_1 R_2$	$Q_1 = (\eta_1, \epsilon_1), Q_2 = (\eta_2, \epsilon_2) \rightarrow$ $Q = Q_1 \cdot Q_2 \leftrightarrow (\eta_1 \eta_2 - \epsilon_1^T \epsilon_2, \eta_1 \epsilon_2 + \eta_2 \epsilon_1 + \epsilon_1 \times \epsilon_2).$

where $*$ denotes the quaternion product operation.

Position of a frame in 3D

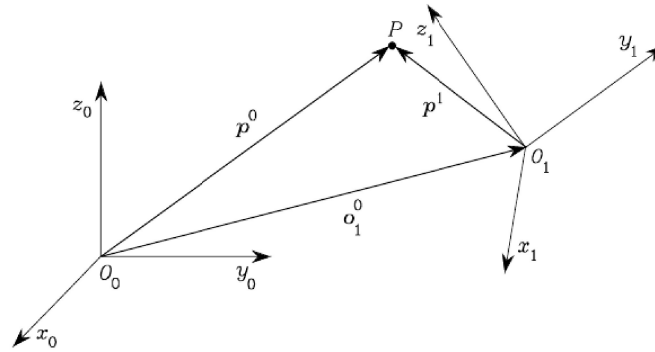


- ▶ At the beginning of this lecture, we have seen that a rigid body is identified by position and rotation of a frame attached to it.
- ▶ Then, we have ignored translation and we have only considered rotation (with the origin of the frames coincident)
- ▶ It is time to go back to the general case
- ▶ it is easy to see that

$$p^0 = o_1^0 + R_1^0 p^1.$$



Direct and inverted coordinate transformation



- The inverted relation of

$$p^0 = o_1^0 + R_1^0 p^1$$

is

$$\begin{aligned} p^1 &= -R_1^{0T} o_1^0 + R_1^{0T} p^0 \\ &= -R_0^1 o_1^0 + R_0^1 p^0 \end{aligned}$$

- This type of representation is tedious.

Homogeneous coordinates

- ▶ We need a compact way to represent both the rotation and the translation
- ▶ The trick is quite simple. Create the *homogenous representation* $\tilde{p}$ of p by appending a fictitious 1 to the vector.

$$\tilde{p} = \begin{bmatrix} p \\ 1 \end{bmatrix}$$

- ▶ We can create the following 4×4 matrix

$$A_1^0 = \begin{bmatrix} R_1^0 & o_1^0 \\ 0^T & 1 \end{bmatrix}$$

which is termed *homogeneous tranformation matrix*.

- ▶ The transformation from frame $O_0 - x_0y_0z_0$ to $O_1 - x_1t_1z_1$ can be written as

$$\tilde{p}^0 = A_1^0 \tilde{p}^1.$$

Homogeneous coordinates

- ▶ A first advantage is that we can write the coordinate transformation in a matrix multiplication form

$$\tilde{p}^0 = A_1^0 \tilde{p}^1.$$

- ▶ The inversion is simply given by

$$\begin{aligned}\tilde{p}^1 &= A_0^1 \tilde{p}^0 \\ &= (A_1^0)^{-1} \tilde{p}^0\end{aligned}$$

- ▶ It is easy to see that

$$A_0^1 = (A_1^0)^{-1} = \begin{bmatrix} R_1^{0T} & -R_1^{0T} o_1^0 \\ 0^T & 1 \end{bmatrix}$$

Homogeneous coordinates

- ▶ Contrary to the rotation matrices, homogeneous transformations are not orthogonal therefore

$$A^{-1} \neq A^T.$$

- ▶ The homogenous formalism allows us to generalise to roto-translations all we have seen for rotations
- ▶ For instance for a sequence of transformation in the current axes, we can write

$$\tilde{p}^0 = A_1^0 A_2^1 \dots A_n^{n-1} \tilde{p}^n.$$

- ▶ The homogenous transformations form a group said special euclidean group SE(3): $SE(3) = \mathbb{R}^3 \times SO(3)$.